



Spring 2026

EMCH792 – Optimal State Estimation

Homework 6

Submission deadline: Wednesday, April 15, 11:59pm EDT

- Submission only through Blackboard, email submissions will not be accepted.
- Make sure you properly upload the file(s) and that you can successfully view your submission on BB. Keep a copy of the submission receipt you get from BB.
- Late submissions will be graded with the policy described in the syllabus.

Homework Instructions

In this homework you will learn how to implement non-linear Kalman Filters in **MATLAB**. All the files that you will need for this homework are in the **ws** folder. You are expected to work inside this folder only. Open **MATLAB** and set the current folder to **ws**, and do the following:

- Start by opening the live script file “HW_6_manual.mlx” using **MATLAB**. This will guide you through the basic implementations of an EKF and a UKF in **MATLAB**.
- Open the file “HW_6.m” and follow the instructions in the file to solve the following problem:

$$x_{k+1} = x_k + 0.5 \cdot dt \cdot r \cdot (ur_k + ul_k) \cdot \cos(\varphi_k)$$

$$y_{k+1} = y_k + 0.5 \cdot dt \cdot r \cdot (ur_k + ul_k) \cdot \sin(\varphi_k)$$

$$\varphi_{k+1} = \varphi_k + dt \cdot \frac{r}{l} \cdot (ur_k - ul_k)$$

$$y_1 = \sqrt{x^2 + y^2}$$

$$y_2 = \tan^{-1}\left(\frac{y}{x}\right)$$

Consider the differential drive system shown above. The state of the system is the x and y position and orientation of the robot $[x, y, \phi]^T$. The process noise covariance is $Q = \text{diag}[0.03, 0.03, 0.20]$. You can measure the polar coordinates of the robot, y_1 with an error with zero mean and variance of 0.0025 and y_2 with an error with zero mean and variance of 0.0025. All the data needed for this homework is in the “HW6_data.mat” file and is loaded at the beginning of the HW_6.m file. The true position and orientation of the robot are x, y and *phi* respectively, the inputs to the system are *ur* and *ul*, and finally y_1 and y_2 are the measurements. For $r=0.1\text{m}$ and $l=0.2\text{m}$ do:

1. Set the initial estimated state to $[0, 0, 0]$, and the initial error covariance to $\text{diag}([0.1, 0.1, 0.01])$. Design an Extended Kalman Filter and simulate the system for all timesteps provided while collecting the estimated state.
2. Set the initial estimated state to $[0, 0, 0]$, and the initial error covariance to $\text{diag}([0.1, 0.1, 0.01])$. Design an Unscented Kalman Filter and simulate the system for all timesteps provided while collecting the estimated state.
3. For both filters, plot and compare the true and estimated states.
4. For both filters, plot and compare the error of the filters and the error bounds.

If you want to write functions to help you with the different parts of this homework you can write them at the bottom of the ‘HW_6.m’ file.

Submission Instructions

Your submission must include the following files:

- “HW_6.m” -> The edited MATLAB script with your solutions

Upload all files on Blackboard in a single submission.

Make sure that the scripts you submit can run. Do not copy the results that appear in the command window in your scripts and make sure that if you add any comments in your scripts that they follow the proper **MATLAB** syntax. **Scripts that do not run will not be graded.**

Grading Rubric

Problem	Points
1	40
2	40
3	10
4	10